

Power Budget Subsystem A2

Team Number: 201

Project Name: EGR 314 Thing

Team Member Names: Michael, Levi, Hafsa, Austin, Isaac, Neel, Seth, Jacob, K

Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	3.3V regulator	LM2575D2T-3.3-ND1	+5V - 35V	1	1000	1000	mA
	ESP32	ESP32-S3-WROOM-1-N4	+1.8 - 3.3V	1	350	350	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Wi-Fi transceiver	(full part number)	+1.8 - 3.3V	1	350	350	mA
						0	mA
						Subtotal	350 mA
						Safety Margin	25%
						Total Current Required on +3.3V Rail	437.5 mA
Regulator	+3.3V low-dropout regulator	LM2575D2T-3.3-ND1	+5V - 35V	1	1000	1000	mA
	Total Remaining Current Available on 3.3V Rail					562.5	mA

<i>External Power Source 1</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	12V AC	+24V	5000	5000	mA
Power Rails Connected to External Power Source 1					0	0	mA
					0	0	mA
	+3.3V low-dropout regulator	LM2575D2T-3.3-ND1	+5V - 35V	1	350	350	mA
Total Remaining Current Available on External Power Source 1						4650	mA

Notes:

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit